

The large-scale dynamic modeling study on the visibility-dependent neural coding of sensory information

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Introduction

1. In research on perceptual awareness, theories, such as the global neuronal workspace theory (GWT) and the information integration theory (IIT), have postulated different neural mechanisms and brain regions underlying the temporal dynamics and processing of stimuli.
2. However, no studies have yet mechanistically modeled and validated these theoretical frameworks.
3. To address this gap, we conducted an EEG-based analysis of neural representations related to the orientations of backward-masked Gabor stimuli. Additionally, we developed a large-scale dynamic model to elucidate the underlying mechanisms of stimulus representations across different states of awareness.

Experimental paradigm

Behavior results (N=27)

A slight change in mask contrast induces blindsight-like behavioral performance.

Visibility dependent coding dynamics (sensor space)

There exists a *switch* in the spatio-temporal dynamics of stimulus representation, indicating the involvement of distinct neural mechanisms.

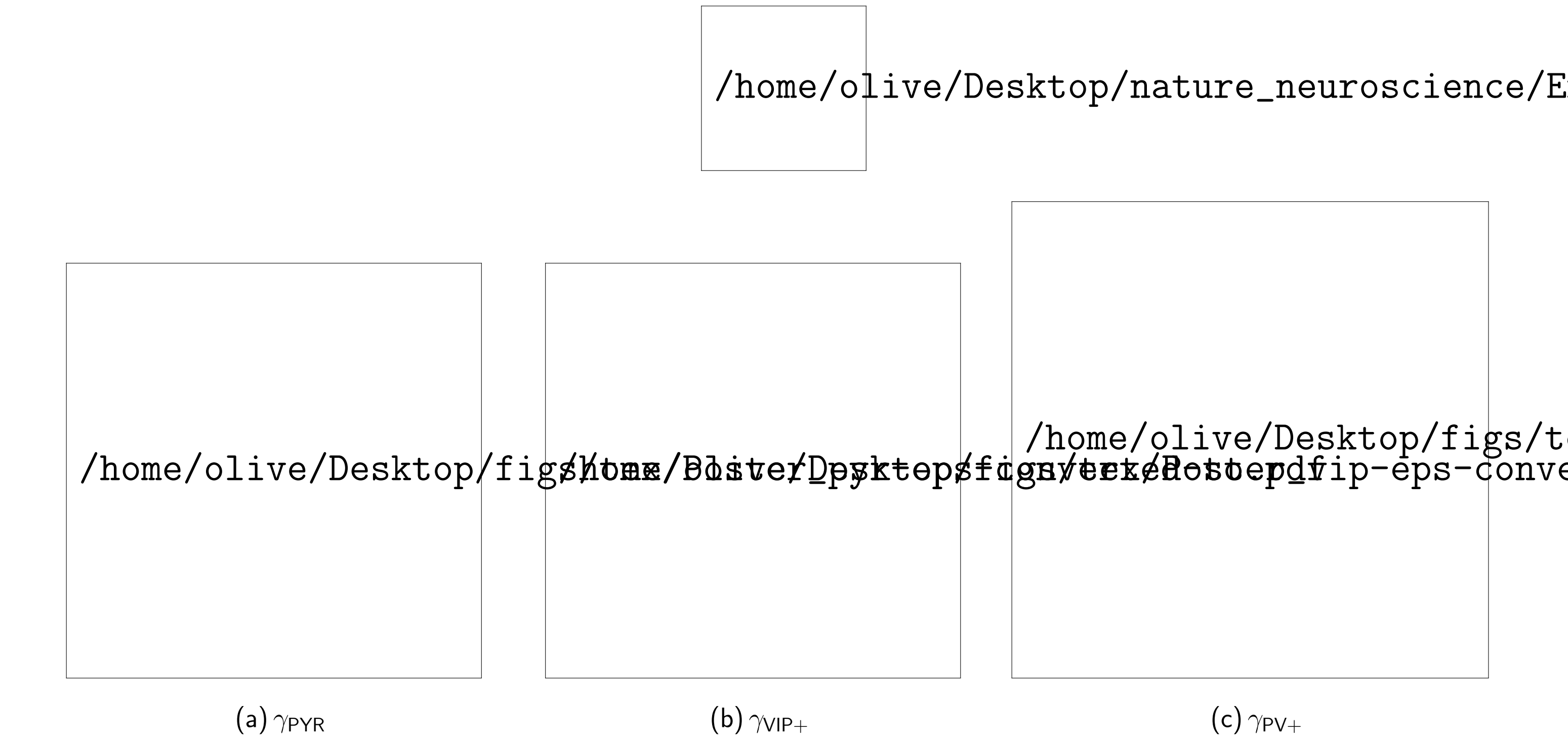
In visible conditions, robust dynamic neural representations in sensory and central areas suggest that this mechanisms may be sufficient to support initial stages of conscious awareness. During invisible conditions, with weak dynamic coding, early stable neural representations of the stimulus emerge from the frontal cortex, implicating a potential role of frontal recurrent neural circuit for conscious awareness.

Visibility dependent coding dynamics (source space)

Source space analysis reveals the initiation of stable neural representations, progressing from frontal to visual cortices, during conditions of perceptual invisibility.

Large-scale dynamic model of consciousness

A five-cell mesoscopic canonical model is proposed for each region within the source space



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